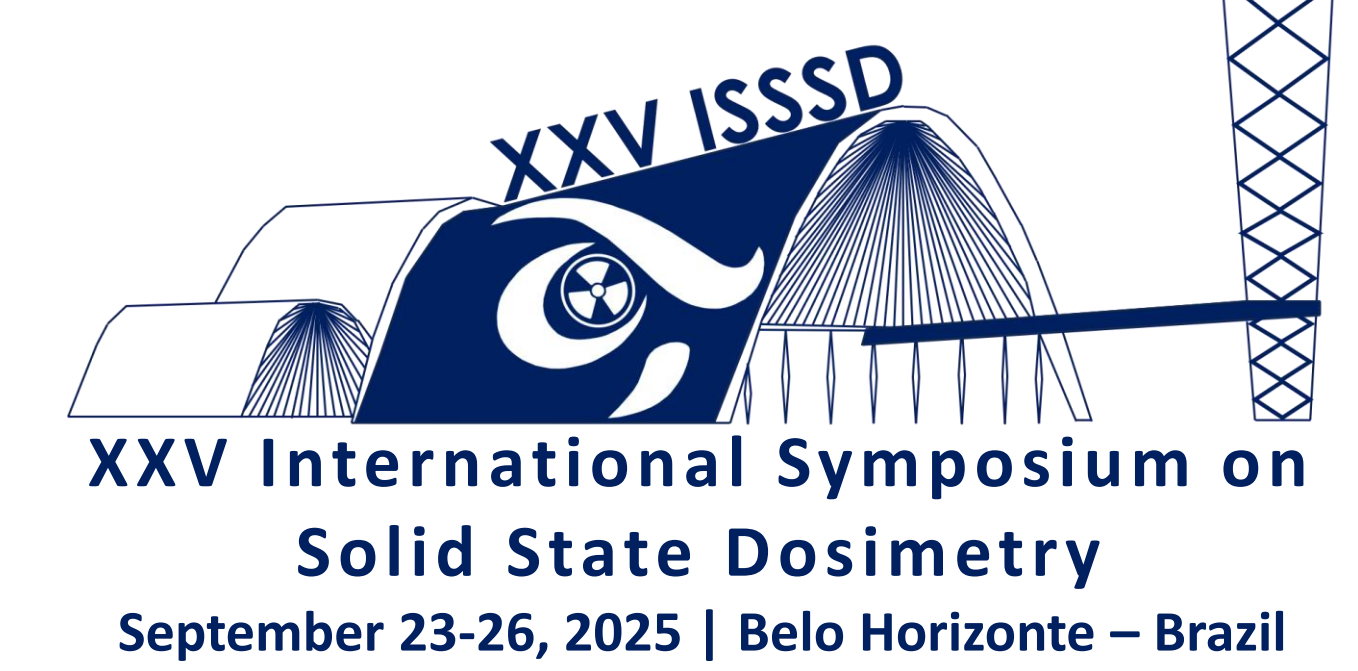


Patient-Specific Dose Optimization in Breast PET/CT: Towards ALARA-Compliant Protocols



A. Sulieman^{*1}, H. Salah², O. Mahgoub³, M. Alkhorayef⁴, , Amjad Alyahyawi⁵, and D. A. Bradley^{6,7}

¹Department of Radiological Sciences, College of Applied Medical Sciences, King Saud Bin Abdulaziz University for Health Sciences, P.O. Box 2477, Al-Ahsa, Al Hofuf 31982, Saudi Arabia

¹ INAYA Medical College, Nuclear Medicine Department, Riyadh, Saudi Arabia

³Department of Radiological Sciences, National Ribat University, Khartoum, Sudan

⁴Department of Radiological Sciences, College of Applied Medical Sciences, King Saud University, Riyadh, Saudi Arabia

⁵Department of Diagnostic Radiology, College of Applied Medical Sciences, University of Hail, Hail, Saudi Arabia

⁶Radiation Dosimetry Research Group, Faculty of Engineering, Multimedia University, Jalan Multimedia, Cyberjaya, Malaysia

⁷School of Mathematics and Physics, University of Surrey, Guildford, UK

*Correspondence to: abdelmoneim_a@yahoo.com

1. INSTRUCTIONS

Introduction:

Breast cancer is the leading cause of cancer-related death in women, with an estimated 2.3 million new cases in 2022, and a rising incidence in many countries due to factors like obesity, alcohol consumption, and lifestyle. Breast PET/CT has emerged as a powerful tool for detecting primary tumors, evaluating treatment response, and identifying metastatic disease in breast cancer patients. However, the combined radiation exposure from ¹⁸F-FDG PET and CT components raises significant concerns, particularly for patients requiring repeated imaging. Current one-size-fits-all dosing protocols often fail to account for critical patient-specific factors such as body habitus, breast density, and clinical indication, potentially leading to unnecessary radiation exposure. The objective of this study is to evaluate adult patient exposure during the PET/CT breast procedure.

Materials and Methods:

The study was conducted at a PET/CT research center in the Almanah Hospital, Eastern Province, Dammam, Saudi Arabia (Figure 1). The hospital is among the country's largest referral hospitals and oncology clinics. The dose from a combined PET/CT (Siemens Biograph mCT Flow) system, known for its precision, and the amount of radiation supplied to the patient were used to determine their exposure (Figure 2). The rotation time is 0.5 s, the table weight capacity is 227 kg, and the size is 78 cm. The maximum output of the CT machine is 140 kVp and 800 mA.

Results:

A total of 25 breast PET-CT scans were analyzed. The mean age of the patients was 52.6 years (SD = 11.47) (Table 1). The mean DLP was 468.66 mGy.cm (SD: 185.77). The mean CTDI was 6.62 mGy (SD: 0.5). The mean radionuclide activity was 308.58 MBq (SD: 14.57) (Table 2).

Conclusions:

The results underscore the potential for further dose reduction through tailored approaches that account for individual patient characteristics such as body habitus and clinical indication.



Figure 1.Almana Hospital, Eastern Province, Dammam, Saudi Arabia

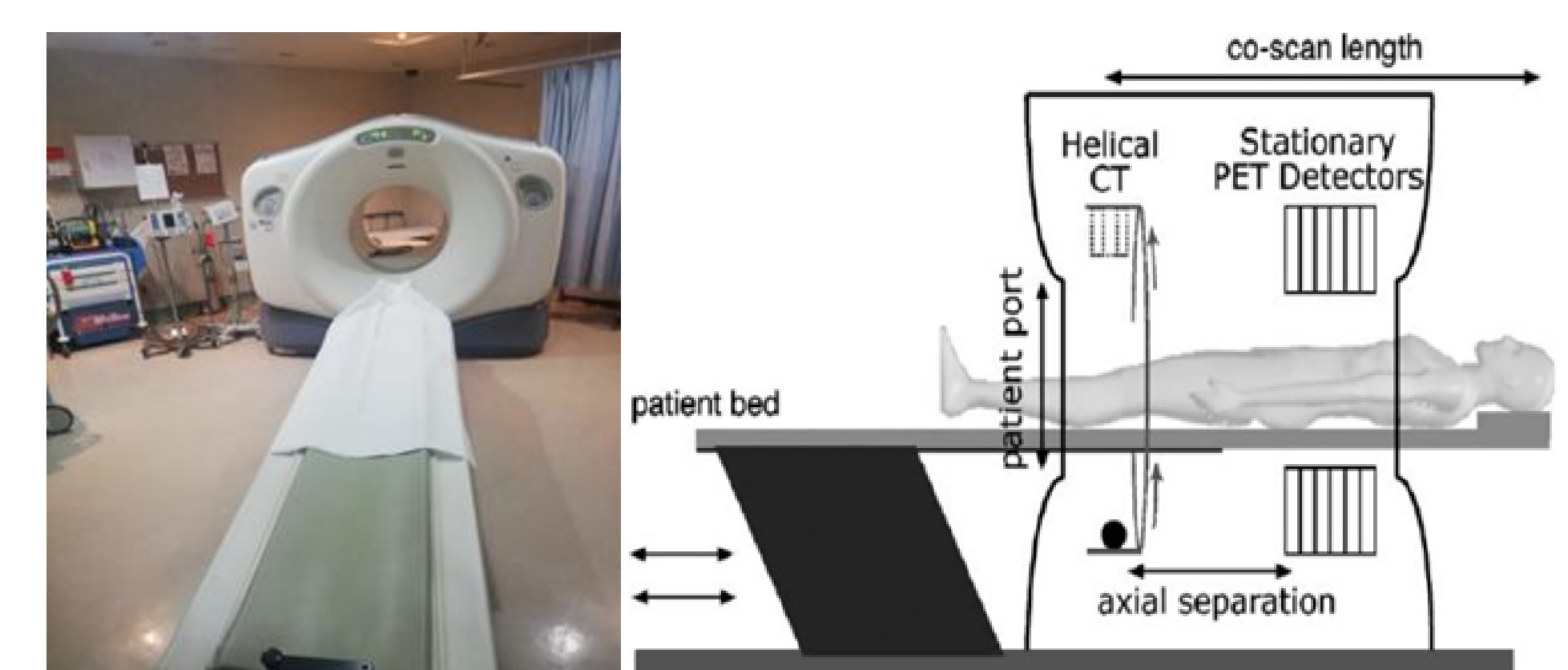


Figure 2.PET CT system

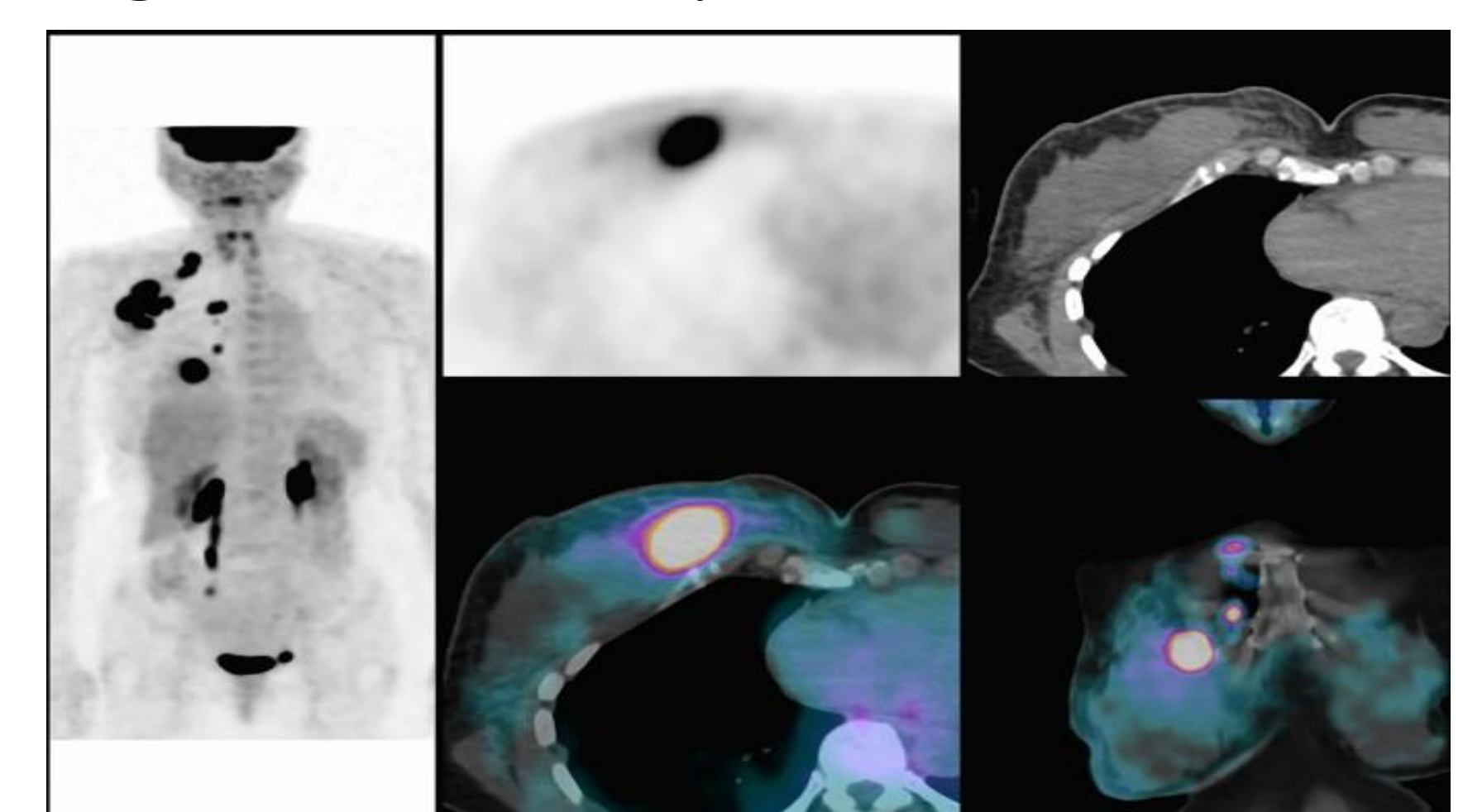


Figure 3. Breast cancer: initial workup and staging with FDG PET/CT

Table 1: Patient mean, standard deviation, minimum and maximum demographic data

Age (y)	Height (cm)	Wight (kg)	BMI (kg/m ²)
52.6±11.4 (34-68)	157±9 (144-170)	63.8±8.1 (47.8-74.4)	25.8±2 (19-29.8)

Table 2: dose Parameters

Activity (MBq)	DLP (mGy.cm)	CTDI (mGy)	Scan Length (mm)	SSDE (mGy)	KVp
308.58±14 (284-321)	468.6±18 5 (138-589)	6.6±0.4 (5.7-6.9)	588.5±357 (246-1154)	7.384± 1.5 (5.4-9.8)	120

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